

GEETHA KRISHNA GURUJU

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EDUCATION

New York University

New York, NY

Master's in Computer Engineering; GPA: 3.96/4.0

Sep 2024 - May 2026

Courses: Machine Learning, Deep Learning, Adv ML, ML for CyberSec, ML Ops & Eng

Jawaharlal Nehru Technological University

Hyderabad, India

Bachelor of Technology, Information Technology; GPA: 9.0/10.0

July 2019 - May 2023

TECHNICAL SKILLS

LLM/AI Platforms: Kubernetes (EKS/on-prem), MLflow, Ray, LLM Ops, Triton, PyTorch

Cloud & DevOps/IaC: AWS, Docker, Terraform, CI/CD, Git, Jenkins, Agile

Big Data Processing: Apache Spark, Scala, Airflow, Kafka, SQL

Programming & Scripting: Python, Java, C/C++, JavaScript

EXPERIENCE

JP Morgan Chase & Co.

Hyderabad, India

Software Engineer | Scala | Python | Spark | Airflow | AWS | Kubernetes

Jun 2023 - Aug 2024

- Built and optimized **distributed ETL pipelines** using Spark and AWS Glue, improving throughput through schema and partitioning strategies and achieving a **7× faster** end-to-end runtime.
- Developed **Airflow ingestion workflows** deployed on **Kubernetes** using Docker and Terraform, with automated validation and monitoring via AWS Lambda for reliable, high-quality data delivery.
- Collaborated with engineering and product teams to translate requirements into scalable solutions, improving system visibility and reducing **AWS operational costs by 60%**.

Software Engineer Intern | Python | ML | NLP | Information Retrieval

Feb 2023 - May 2023

- Researched large-scale data comparison and ranking, focusing on Locality Sensitive Hashing algorithms.
- Built ML-based Learn to Rank model to effectively reduce false positives in enterprise screening tasks.

Summer Intern | SQL | Robotic Process Automation | WinAutomation

May 2022 - Jul 2022

- Developed and optimized RPA workflows using WinAutomation for operations engineering teams.

New York University

New York, NY

Teaching Assistant | CSCI-GA 3033: Intro to LLM-based Generative AI Systems

Sep 2025 - Present

- Guided students through **assignments, and projects** involving LLM fine-tuning, inference optimization, and model evaluation using PyTorch.
- Supported coursework on **LLM application development**, including prompt engineering, LangChain-based pipelines, RAG workflows, and tool-augmented LLM and agentic AI frameworks.

Tutor | CSCI-UA 0472: Artificial Intelligence

Sep 2025 - Present

- Tutored students on ML, search, probabilistic models, logic, and NLP programming assignments.

PROJECTS

OncoReason: Structuring Clinical Reasoning in LLMs | LLM, GRPO, CoT 📄 Jan 2025 - Aug 2025

- Built a reasoning-aligned framework for treatment outcome prediction on MSK-CHORD by integrating **Chain-of-Thought prompting** with **GRPO fine-tuning** on open-source LLMs (LLaMA3, Med42).
- Jointly modeled survival status, duration, and rationales (**F1 = 0.84**, **BLEU = 98.15**) and developed a reward-based RL method to improve semantic fidelity and clinical interpretability. Published on arXiv.

MLOps Platform for Medical QA System | Kubernetes, Triton, Ray, MLflow 📄 Jan 2025 - May 2025

- Designed an end-to-end ML system with modular ingestion, data validation, schema enforcement, and versioned dataset management for reliable retraining cycles.
- Implemented distributed training pipelines using **Ray** and PyTorch Lightning, integrated with **MLflow** and MinIO for experiment tracking, artifact storage, and reproducible model development.
- Deployed optimized inference services on **Kubernetes** using **Triton Inference Server**, incorporating autoscaling, load balancing, drift monitoring, and health checks with real-time dashboards in Streamlit, reinforcing deployment concepts from course labs.

CERTIFICATIONS & HONORS

AWS Certified Cloud Practitioner | NASA Space Apps'22 Winner | JPMC Code for Good'22 2nd Place